Project | Sustainability Impact Analysis for Intel

INTRODUCTION: As you learned listening in on the strategy meeting with Dr. Alvarez and Intel's Sustainability Team, Intel is committed to reducing its carbon footprint and improving the sustainability of its devices – not just during manufacturing, but throughout the entire lifecycle.

A key part of this effort is their repurposing programs, which play a central role in achieving these sustainability goals. Repurposing and recycling programs aim to reduce e-waste, energy consumption, and CO_2 emissions by extending the life of existing devices, and thus reducing the need for new device manufacturing. Like Michael Campbell said: the average household in the US has anywhere from 3–5 PCs devices, tablets, notebooks, desktops that are perfectly functional, but not being used!

One challenge Intel faces is determining which devices in its repurposing program should be prioritized for the maximum environmental benefit. That's where data analysis comes in! To help with this, Intel gathered data on each device repurposed or recycled in 2024.

Your task is to evaluate the effectiveness of Intel's current repurposing strategy and provide a data-driven recommendation to help guide the program's direction and optimize sustainability efforts.

HOW IT WORKS: Follow the prompts in the questions below to investigate the data. Post your answers in the provided boxes: the **yellow boxes** for the queries you write and **blue boxes** for your text-based analysis. Once you're done, you'll submit your ***completed*** .pdf file to HQ for feedback from The Accelerator Team.

SQL App: Here's the link to our specialized SQL app, where you'll write your SQL queries and interact with the data.

NOTE: The dataset you are working with is designed for The Global Career Accelerator to reflect the key characteristics and structure of Intel's real data, while protecting their confidentiality and proprietary information. Be aware that any conclusions or results derived from this dataset should be viewed as hypothetical and for illustrative purposes only.

— Data Set **Descriptions**

In this project you'll query 2 different datasets, `intel.device_data` and `intel.impact_data`, that you will join together for your analysis. Here you'll find the data dictionary for each dataset.

`intel.device_data`

- `device_id`: Unique identifier for each repurposed device
- `device_type`: Type of device, values are either "Laptop" or "Desktop"
- `model_year`: The year the device was manufactured (e.g., 2018, 2019, etc.)

`intel.impact_data`

- `impact_id`: Unique identifier for the repurposed device's impact record (e.g., "LP20NA141592")
- `device_id`: Unique identifier linking the impact record to a specific device in the `intel.device_data` table
- `usage_purpose`: The specific purpose for which the device is being repurposed, values are Education & Digital Literacy, Corporate & Enterprise, Government & Public Sector, Environmental Sustainability Programs, and Social Impact & Non-Profit
- `power_consumption`: Power consumption of the device in watts (W) when in use (e.g., 50W, 75W)
- `energy_savings_yr`: Estimated energy savings per device per year when repurposed compared to a new device, measured in kilowatt-hours (kWh)
- `co2_saved_kg_yr`: Estimated CO2 emissions saved per device per year from manufacturing a new device, measured in kilograms (kg).
- `recycling_rate`: The percentage of the device that is recyclable (e.g., 80%, 90%).
- `region`: The geographical region where the device was repurposed, values are "North America", "Europe", and "Asia"

— **Task 1:** Organizing and Understanding the Data

We'll start by **joining** the device data with the impact data, allowing for a comprehensive analysis of device types, model years, repurpose regions, and energy savings in one dataset.

A. Simply write a query that returns all of the columns from both tables, joining the two on the `device_id` column. Be sure to choose the appropriate join so that all relevant

data is included in your result. **Note:** your query will have more than 150,000 rows (the max display for SQLPad!)

(paste your query below 👇)

```
SELECT
    *
FROM intel.device_data device
LEFT JOIN intel.impact_data input
    ON device.device_id = input.device_id;
```

B. To your joined dataset, add a new column called `device_age` calculated by subtracting the model_year from 2024. Paste your query below and double check that the values in your new column make sense. For example, a 2019 device should be 5 years old.

(paste your query below 👇)

```
SELECT
    device.*,
    input.*,
    2024 - device.model_year AS device_age
FROM intel.device_data device
LEFT JOIN intel.impact_data input
    ON device.device_id = input.device_id;
```

C. Order your joined data by model_year (oldest to newest). Do you notice more older (5+ years) or newer (under 5 years) devices being repurposed? What might that indicate?

(write your **answer** below 👇)

I noticed from the data that there are more older devices that are 5+ years old being repurposed than newer ones. This might be a sign that

Intel is trying to reuse older computers so they don't get thrown away or turned into trash.

D. Bucketing the `device_age` will allow us to analyze trends and patterns in energy savings and CO_2 reductions more effectively than using individual ages. Use a `CASE WHEN` clause to add one more column, called `device_age_bucket`, to your data, that is based on the device_age:

- `WHEN` the `device_age` is less than or equal to 3, `device_age_bucket` should be "newer"
- `WHEN` the `device_age` is greater than 3 but less than or equal to 6, `device_age_bucket` should be "mid-age"
- `WHEN` the `device_age` is greater than 6, `device_age_bucket` should be "older"

HINT: Instead of using e.g. `device_age <= 3`, you need to reference the calculation directly: `2024 - d.model_year <= 3`.

Double check that the values in your new column make sense! For example, a 2019 device should be characterized as "mid-age".

(paste your query below 👇)

```
SELECT
  device.*,
  input.*,
  2024 - device.model_year AS device_age,
  CASE
    WHEN 2024 - device.model_year <= 3 THEN 'newer'
    WHEN 2024 - device.model_year <= 6 THEN 'mid-age'
    ELSE 'older'
  END AS device_age_bucket
FROM
  intel.device_data device
  LEFT JOIN intel.impact_data input ON device.device_id
= input.device_id
```

```
ORDER BY
  device.model_year ASC;
```

— **Task 2:** Key Insights

Now it's time to analyze the overall impact of Intel's repurposing program. You will use your final query from **Task 1** together with the `WITH` keyword for the remainder of this Project as you aggregate and analyze the data you've organized and prepped. For a refresher, rewatch "🍿The `WITH` Keyword" in SkillBuilder 6.

A. What is the total number of devices Intel repurposed in 2024?

HINT: The dataset **is** representing all devices repurposed in 2024! You just need to COUNT all the rows in your joined data from Task 1!

(write your **answer** below 👇)

The total number of devices Intel repurposed in 2024 was 601,740.

B. Write a query that returns the total number of devices repurposed, the average age of repurposed devices in 2024, the average estimated energy savings (kWh) from repurposed devices per year, and the total CO_2 emissions saved (in tons) from repurposed devices.

Note: CO_2 emissions are typically measured in tons. Since `CO₂_saved_kg_yr` is measured in kg, divide the `SUM(CO₂_saved_kg_yr)` by 1000 to report the total CO_2 emissions saved in tons.

(paste your query below 👇)

```
WITH intel_data AS (
  SELECT
    device.*,
    input.*,
    2024 - device.model_year AS device_age,
```

```
    CASE
      WHEN 2024 - device.model_year <= 3 THEN 'newer'
      WHEN 2024 - device.model_year <= 6 THEN 'mid-age'
      ELSE 'older'
    END AS device_age_bucket
  FROM
    intel.device_data device
    LEFT JOIN intel.impact_data input ON
device.device_id = input.device_id
)
SELECT
  COUNT(*) AS total_devices,
  AVG(device_age) AS avg_device_age,
  AVG(energy_savings_yr) AS avg_energy_savings_kwh,
  SUM(co2_saved_kg_yr) / 1000.0 AS total_co2_saved_tons
FROM
  intel_data;
```

C. Now that you have calculated the average estimated energy savings (kWh) and CO_2 emissions saved (tons), use ChatGPT to help put these numbers into perspective.

Try this prompt: I found that each repurposed device saves approximately of XXX kWh of energy per year and Intel's repurposing program saved XXX tons of CO_2 emissions in one year. Help me understand the significance of these numbers. How would this compare to the energy consumption of a small city or the amount of CO_2 produced by cars? What is the environmental impact of these savings?

What comparisons did you find most impactful in terms of scale? Summarize how much energy and CO_2 emissions were saved and how it compares to something familiar, like powering households or reducing car emissions.

(write your **answer** below 👇)

I noticed that Intel repurposed devices saved about 25.7 kWh of energy per device each year, and about 6768.4 tons of CO2 total. I feel like it's like providing electricity to many homes for a year, or simply removing cars from the road, which can prevent a lot of pollution. I see that these savings show that reusing older devices instead of getting rid of them immediately can make a huge difference for the environment, while reducing waste and saving energy in the most effective way possible.

– **Task 3:** Identifying Trends & Maximizing Sustainability

By grouping our data in different ways, we can uncover patterns in energy savings and CO_2 reductions. These insights will help us determine which categories of devices contribute the most to sustainability efforts and where Intel should focus its repurposing strategy for maximum impact.

A. Write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), grouped by device_type.

Note (again): You'll need to divide `AVG(CO₂_saved_kg_yr)` by 1000 to report the average CO_2 emissions saved in tons.

(paste your query below 👇)

```
WITH intel_data AS (
    SELECT
        device.*,
        input.*,
        2024 - device.model_year AS device_age,
        CASE
            WHEN 2024 - device.model_year <= 3 THEN
'newer'
            WHEN 2024 - device.model_year <= 6 THEN
'mid-age'
            ELSE 'older'
        END AS device_age_bucket
    FROM intel.device_data device
```

```
    LEFT JOIN intel.impact_data input
        ON device.device_id = input.device_id
)
SELECT
    device_type,
    COUNT(*) AS total_devices,
    AVG(energy_savings_yr) AS avg_energy_savings_kwh,
    AVG(co2_saved_kg_yr) / 1000.0 AS avg_co2_saved_tons
FROM intel_data
GROUP BY device_type
ORDER BY total_devices DESC;
```

B. Based on the results, which device type contributes the most to energy savings and CO_2 reduction? Why might that be the case?

Hint: Don't forget you can use ChatGPT as your Teammate to help think through your response!

(write your **answer** below 👇)

Laptops contribute the most to energy savings and CO2 reduction. I assume the case is that more laptops are being repurposed than desktops. Even though each device saves almost the same amount of energy, having so many laptops makes the overall environmental impact greater.

C. Write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), now grouped by `device_age_bucket`.

(paste your query below 👇)

```
WITH intel_data AS (
  SELECT
```

```
    device.device_id,
    2024 - device.model_year AS device_age,
    CASE
      WHEN 2024 - device.model_year <= 3 THEN 'newer'
      WHEN 2024 - device.model_year > 3
      AND 2024 - device.model_year <= 6 THEN 'mid-age'
      ELSE 'older'
    END AS device_age_bucket,
    input.energy_savings_yr,
    input.co2_saved_kg_yr
  FROM
    intel.device_data device
    LEFT JOIN intel.impact_data input ON
device.device_id = input.device_id
)
SELECT
  device_age_bucket,
  COUNT(*) AS total_devices,
  AVG(energy_savings_yr) AS avg_energy_savings_kwh,
  AVG(co2_saved_kg_yr) / 1000.0 AS avg_co2_saved_tons
FROM
  intel_data
GROUP BY
  device_age_bucket
ORDER BY
  total_devices DESC;
```

D. Based on the result of your query, what do you notice about the relationship between device age and the number of devices repurposed versus the average energy saved?

(write your **answer** below 👇)

The newer devices are being repurposed more now; however, the older devices save more energy and CO2 per device. This shows that

repurposing older devices is crucial and still has a bigger impact towards the environment, even if there are fewer of them.

E. Finally, write a query that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), now grouped by region.

(paste your query below 👇)

```
WITH intel_data AS (
  SELECT
    device.*,
    input.*,
    2024 - device.model_year AS device_age,
    CASE
      WHEN 2024 - device.model_year <= 3 THEN 'newer'
      WHEN 2024 - device.model_year <= 6 THEN 'mid-age'
      ELSE 'older'
    END AS device_age_bucket
  FROM
    intel.device_data device
    LEFT JOIN intel.impact_data input ON
device.device_id = input.device_id
)
SELECT
  region,
  COUNT(*) AS total_devices,
  AVG(energy_savings_yr) AS avg_energy_savings_kwh,
  AVG(co2_saved_kg_yr) / 1000.0 AS avg_co2_saved_tons
FROM
  intel_data
GROUP BY
  region
ORDER BY
  total_devices DESC;
```

F. How does the carbon intensity of electricity in each region impact the total CO_2 savings from repurposed devices? Are there regions where repurposing leads to significantly higher environmental benefits? Why might that be?

(write your **answer** below 👇)

The amount of CO2 saved basically depends on how much carbon is in the electricity for each region. Although North America has the most devices repurposed, Asia saves the most CO2 per device due to its electricity being higher in carbon. Europe has the fewest CO2 savings because it has cleaner electricity. This data shows that reusing devices in areas that use high-carbon electricity can actually result in positive impacts on the environment.

— **Task 4:** Data-Driven Recommendations

Using the findings from this analysis, we need to summarize key takeaways and develop actionable recommendations for Intel. Remember: the goal is to refine Intel's repurposing strategy to maximize energy savings and CO_2 reductions while ensuring the most effective use of resources.

A. Based on your analysis of the repurposed devices (including energy savings, CO_2 emissions, and device age), write **four** key takeaways in succinct sentences/bullets that summarize the most important patterns and insights from the data. These should be specific, concise, and focused on the implications of repurposing newer versus older devices.

(write your **answer** below 👇)

- The older devices save more energy and CO2 per device than newer devices, even though less is repurposed.
- Laptops are repurposed more often and definitely contribute the most to the total energy and CO2 savings
- Regions with high carbon emissions, such as Asia, benefit more from repurposing devices than regions with cleaner electricity, like Europe.

- Reusing devices rather than getting rid of them helps reduce waste, saves energy, and lowers CO2 emissions.

B. Based on your four key takeaways and ChatGPT as your teammate, write a recommendation for Intel on how to improve the repurposing program. Your recommendation should include a clear action or strategy for Intel based on the data and a data-driven justification for why this approach would maximize energy savings and CO_2 reductions.

(write your **answer** below 👇)

I feel like Intel should focus on repurposing older laptops because they save the most energy and CO2 per device. These devices tend to be usable for a long amount of time after repurposing. Its preferably they prioritise repurposing devices in regions with high carbon electricity, such as Asia, to maximize the environmental benefits. Lastly, this strategy will ensure the greatest overall impact on reducing emissions and waste while saving so much energy efficiently.

C. Briefly reflect on how ChatGPT's suggestions influenced your recommendation. Did it help you see something you hadn't considered? What parts of your recommendation were improved based on its response?

(write your **answer** below 👇)

It helped me understand the trends in the data as well as how focusing closely on device age and the region affects sustainability results, rather than looking at the number of devices being repurposed. It also had me with more thoughts about which device types and regions to focus more on, and that made my recommendation stronger and more data-driven.

— **LevelUp:** Optimizing Repurposing Strategy for Maximum Impact

Now that you've gained insights into the energy savings and CO_2 reductions across different device types and regions, let's use this data to optimize Intel's repurposing strategy for maximum environmental benefit.

A. Add to your final query of Task 3 that returns the total number of devices, the average energy savings, and the average CO_2 emissions saved (in tons), grouped by region, **the percentage** of the total energy savings and CO_2 reductions contributed by each device type within each region.

HINT: To calculate the percentage of the total energy savings, use this formula:
`Total energy savings for the device type / Total energy savings for the region) * 100`
You'll use a similar one for the percentage of the total CO_2 reductions.

Try this prompt: What's the best way to calculate the percentage of CO_2 reductions contributed by each device type in each region?

(paste your query below 👇)

```
WITH intel_data AS (
  SELECT
    device.*,
    input.*,
    2024 - device.model_year AS device_age,
    CASE
      WHEN 2024 - device.model_year <= 3 THEN 'newer'
      WHEN 2024 - device.model_year <= 6 THEN 'mid-age'
      ELSE 'older'
    END AS device_age_bucket
  FROM
    intel.device_data device
    LEFT JOIN intel.impact_data input ON
device.device_id = input.device_id
```

```
),
region_totals AS (
  SELECT
    region,
    SUM(energy_savings_yr) AS
total_energy_savings_region,
    SUM(co2_saved_kg_yr) AS total_co2_saved_region
  FROM
    intel_data
  GROUP BY
    region
)
SELECT
  intel_data.region,
  intel_data.device_type,
  COUNT(*) AS total_devices,
  AVG(intel_data.energy_savings_yr) AS
avg_energy_savings_kwh,
  AVG(intel_data.co2_saved_kg_yr) / 1000.0 AS
avg_co2_saved_tons,
  (
    SUM(intel_data.energy_savings_yr) /
region_totals.total_energy_savings_region
  ) * 100 AS pct_energy_savings_in_region,
  (
    SUM(intel_data.co2_saved_kg_yr) /
region_totals.total_co2_saved_region
  ) * 100 AS pct_co2_saved_in_region
FROM
  intel_data
  JOIN region_totals ON intel_data.region =
region_totals.region
GROUP BY
  intel_data.region,
  intel_data.device_type,
  region_totals.total_energy_savings_region,
  region_totals.total_co2_saved_region
```

```
ORDER BY
  intel_data.region,
  pct_co2_saved_in_region DESC;
```

B. Based on the results of your query, analyze the data to answer:

- Which device types in which regions contribute the most energy savings and CO_2 reductions relative to their numbers?
- How can this analysis help Intel prioritize specific device types in certain regions to maximize environmental benefits?

(write your **answer** below 👇)

Laptops in Asia contribute the most to CO2 reductions compared to other devices, because the electricity there is higher in carbon, so each repurposed device saves more emissions. This shows Intel should focus on repurposing laptops with higher carbon electricity in order for positive outcomes and more benefits.

C. In addition to focusing on sustainability, imagine Intel needs to optimize for cost-effectiveness in their repurposing program. How might you adjust your query to incorporate cost data (e.g., cost per repurposed device)? What strategies could Intel use to balance sustainability goals with cost constraints?

(write your **answer** below 👇)

First, Intel could add a column for the cost of repurposing each device and calculate the total cost along with energy and CO2 savings. However, I feel like this would help Intel choose devices that provide the greatest sustainability benefits while also keeping the cost low and using their money wisely

– Evaluation Rubric

Unlike your Milestones that were evaluated largely based on your effort, the evaluation of your Portfolio Project will follow traditional evaluation methods, with tasks assessed for correctness and assigned point values accordingly.

Partial credit will be given where parts of this task are correct, even if other parts are incorrect or incomplete.

Task title	Max points
Task 1: Organizing and Understanding the Data	40
Task 2: Key Insights	25
Task 3: Identifying Trends & Maximizing Sustainability	60
Task 4: Data-Driven Recommendations	75
TOTAL POINTS:	200
LevelUp	
Optimizing Repurposing Strategy for Maximum Impact	20